

§3.4

Subtraction in Other Bases

The same two methods, plus one new idea: the anti-dot

Today's Goal

After completing this section, you should be able to:

- Subtract numbers in a base other than ten using the standard algorithm
- Use the Exploding Dots model, including anti-dots, to subtract numbers in a base other than ten

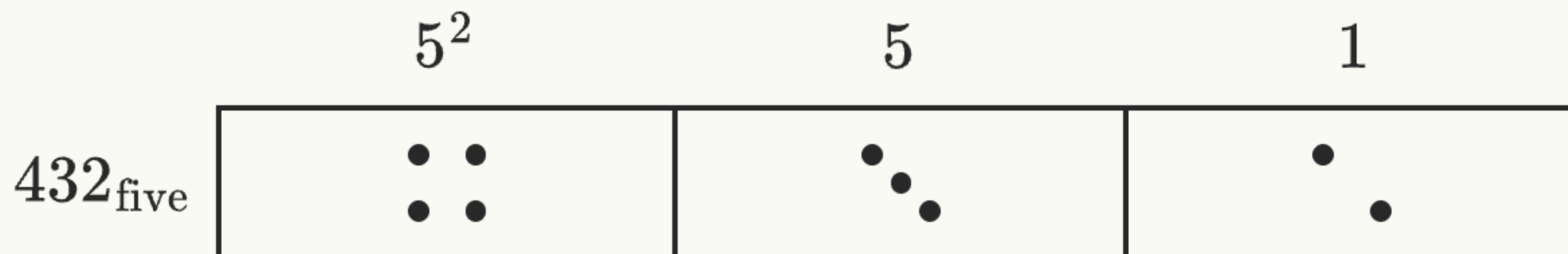
Learning Target 9

LT.9 Different Number Bases

I can convert numbers from base ten to base n and vice versa for any whole number $n > 1$. I can add, subtract, multiply, and divide in different number bases.

Which One Needs More Work?

Evaluate $432_{\text{five}} - 20_{\text{five}}$. Then evaluate $432_{\text{five}} - 24_{\text{five}}$.



For $432_{\text{five}} - 20_{\text{five}}$, we remove 2 dots from the fives position. This leaves us with 412_{five} .

For $432_{\text{five}} - 24_{\text{five}}$, we also have to remove 4 dots from the ones position, and it holds only 2.

Regrouping Along the Way

Evaluate $432_{\text{five}} - 24_{\text{five}}$ using the standard algorithm.

	5^2	5	1
432_{five}			
After Regrouping			
After Subtracting			

This leaves us with 403_{five} . So $432_{\text{five}} - 24_{\text{five}} = 403_{\text{five}}$.

The Same Work Without Dots

$$\begin{array}{r}
 2 12 \\
 4 \cancel{3} \cancel{2} \\
 2 4 \\
 \hline
 4 0 3
 \end{array}$$

Taking 1 from the fives position gives the ones $2 + 5 = 7$, written 12 in base-5.

In base-5, $12 - 4$ is 3, not 8.

Your Turn

Evaluate $2201_{\text{four}} - 312_{\text{four}}$ using the standard algorithm.

		11	3	
	1	1	10	11
	2	2	0	1
—		3	1	2
	1	2	2	3

Taking 1 from the sixteens position gives the fours $0 + 4 = 4$, written 10 in base-4.

Taking 1 from the fours position gives the ones $1 + 4 = 5$, written 11 in base-4.

Taking 1 from the sixty-fours position gives the sixteens $1 + 4 = 5$, written 11 in base-4.

So $2201_{\text{four}} - 312_{\text{four}} = 1223_{\text{four}}$.

Dots and Anti-Dots

In the Exploding Dots model, we deal with regrouping last. So how do we remove 4 dots from a box that holds only 2?

In the Exploding Dots model, a dot represents the number 1. An anti-dot represents the number -1 .

● **Dot:** 1

○ **Anti-dot:** -1

When a dot and an anti-dot meet, they annihilate each other (because $1 + (-1) = 0$).

Adding Anti-Dots

Evaluate $432_{\text{five}} - 24_{\text{five}}$ using the Exploding Dots model.

Instead of subtracting 24_{five} , we will add -24_{five} .

$1 \leftarrow 5$

			
+			
After Annihilating			
			

We still have 2 anti-dots in the ones position.

Regrouping Last

Evaluate $432_{\text{five}} - 24_{\text{five}}$ using the Exploding Dots model.

So we regroup one of the dots in the fives position to become 5 dots in the ones position and annihilate more pairs of dots and anti-dots.

$1 \leftarrow 5$

After Annihilating

		
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After Regrouping

		
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So $432_{\text{five}} - 24_{\text{five}} = 403_{\text{five}}$, the same answer as Example 1.

Annihilate in Every Position

Evaluate $3102_{\text{four}} - 1231_{\text{four}}$ using the Exploding Dots model.

We begin by setting up 3102_{four} and adding -1231_{four} .



We still have anti-dots in the fours and sixteens positions.

Left to Right, One Round at a Time

Evaluate $3102_{\text{four}} - 1231_{\text{four}}$ using the Exploding Dots model.

$1 \leftarrow 4$

After Annihilating

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After Regrouping 1

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After Regrouping 2

We are left with 1211_{four} . So $3102_{\text{four}} - 1231_{\text{four}} = 1211_{\text{four}}$.

Your Turn on the Machine

Evaluate $5603_{\text{eight}} - 1025_{\text{eight}}$ using the Exploding Dots model.

A negative number in a box means that box holds anti-dots.

				$1 \leftarrow 8$
+	5	6	0	3
	-1	0	-2	-5
After Annihilating	4	6	-2	-2
After Regrouping 1	4	5	6	-2
After Regrouping 2	4	5	5	6

So $5603_{\text{eight}} - 1025_{\text{eight}} = 4556_{\text{eight}}$.

When Does Each Method Regroup?

Examples 1 and 2 found the same difference. When did each one regroup?

In the standard algorithm, regrouping takes place along the way.

In contrast, regrouping is reserved for the end in the Exploding Dots model.